Instructions: Upload the `tex` file for this document to your Overleaf project. All solutions shall be completed on this packet by typing your solution in LaTeX in the space provided. All solutions should be detailed and should clearly demonstrate the process by which you arrived at the answer. Submit the compiled `pdf` file to D2L by 11:59 PM on the due date below. Submit only a single pdf file of your entire packet. The question will also ask you to make calculations in Python. Upload any Python files to the appropriate folder in your GitHub repository. In this folder, a `py` file is to be submitted for each problem such that when the `py` file is executed, the output (as presented in Python) is the solution to the problem. Academic dishonesty will not be tolerated.

Phase 4 – Homework

MATH 210 – Mathematical Computing and Typesetting

Due: Friday, May 1 by 11:59 PM on D2L

Solutions by Brooks Emerick

Remember to begin each `py` file with `import numpy as np` and `import matplotlib.pyplot as plt`. For this assignment you will also need `from scipy.integrate import solve_ivp`.

1. Create a file that computes the numerical solution to the following initial value problem

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{M}\right)\left(\frac{y}{m} - 1\right) \qquad y(0) = y_0,$$

for parameter values $k > 0$, $m > 0$, and $M > 0$ and an initial value y_0. Your file should plot the solution to the equation with at least three solutions on the same graph, one with $y_0 < m$, $m < y_0 < M$, and one with $y_0 > M$. In the space below, include the Python figure with a caption. In the caption, you must say which equilibrium states are attracting and which are repelling.

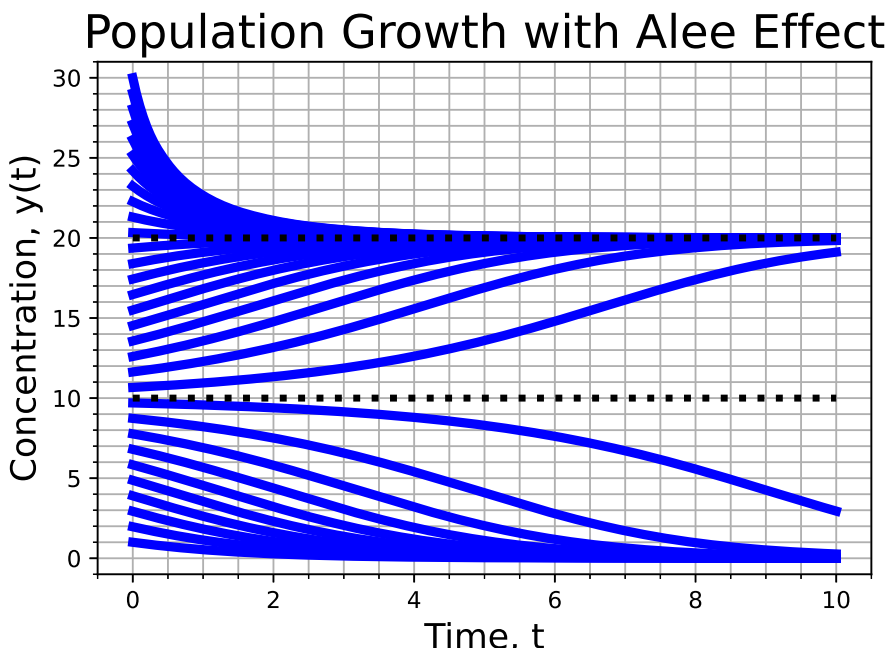


Figure 1: A series of trajectories are shown with starting population values that span three different regimes: $0 < y_0 < m$, $m < y_0 < M$, and $y_0 > M$. Here, we can see that the equilibrium solutions $y^* = 0$ and $y^* = M$ are attracting, while the equilibrium state $y^* = m$ is repelling.

2. Consider the following system of that describes a predator-prey dynamic:

$$\frac{dx}{dt} = Rx - 4x^2 - 3xy$$
$$\frac{dy}{dt} = -2y + xy$$

where $R > 0$ is a real-valued parameter. In this model, which population is the prey and which is the predator? Briefly justify your answer. Determine a value of R that makes both populations coexist. Plot the vector field with the phase plane solution as well. Include the two figures below. Describe in a sentence or two what happens to each population as $t \to \infty$.

We can rewrite the differential equation system in the following form:

$$\frac{dx}{dt} = Rx\left(1 - \frac{x}{R/4}\right) - 3xy$$
$$\frac{dy}{dt} = -2y + xy.$$

Written this way, we can see that the x population experiences logistic growth to the carrying capacity $R/4$ in the absence of the y population. The y population experiences exponential decay in the absence of x. Therefore, it makes sense that the x population is the prey since the interaction term is negative causing a decay if y is nonzero. This makes y the predator since the interaction term is positive and it only grows when x is nonzero.

If coexistence occurs, we see that a nonzero equilibrium of $x^* = 2$ satisfies the second equation. Plugging this value into the first equation yields

$$y^* = \frac{R-8}{3},$$

which means $R > 8$ for the y population to have a positive equilibrium value. Figure 2 shows the phase plane and a time series plot for $R = 60$. As $t \to \infty$, $(x, y) \to (2, 52/3)$, which means both populations approach a positive equilibrium value, i.e. they coexist. We note that x does not achieve its carrying capacity due to the presence of the predator population.

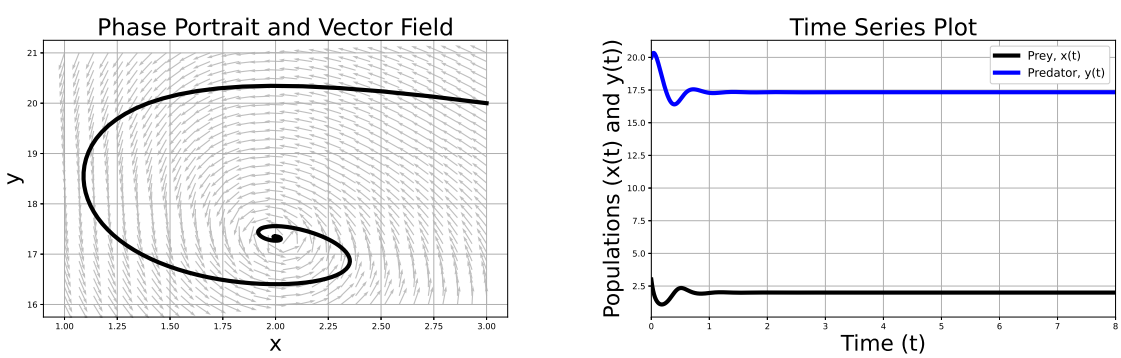
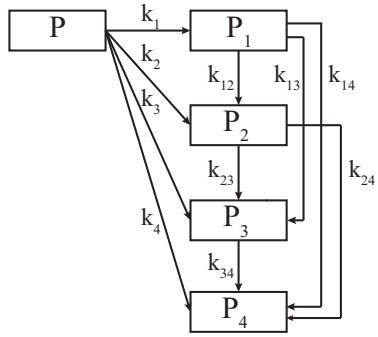


Figure 2: (Left) Phase plane plot with a trajectory and vector field. (Right) Time-series plot for the same trajectory that shows both populations.

3. The following schematic represents a chemical process in which a parent peptide (denoted by P) breaks down into four smaller components (denoted by P_i for $i = 1, 2, 3, 4$). The larger components can also break down into smaller components, where P_4 is the smallest component. Here, assume each "bin" is a function of time and measures the concentration of each component in the solution.



Assume that the initial condition contains only the parent peptide, $P(0) = 1$, and all other components are not yet formed: $P_i(0) = 0$ for $i = 1, 2, 3, 4$. Set up and write down the linear differential equation model that simulates this process. Assume each reaction is constant and positive. Create a file that solves this model and outputs the solution when $k_1 = .09$, $k_2 = .8$, $k_3 = .07$, $k_4 = .6$, $k_{12} = .05$, $k_{13} = .5$, $k_{14} = .06$, $k_{23} = .7$, $k_{24} = .08$, and $k_{34} = .9$. Include the figure below in your LaTeX write-up and describe in the caption what happens in this system as $t \to \infty$.

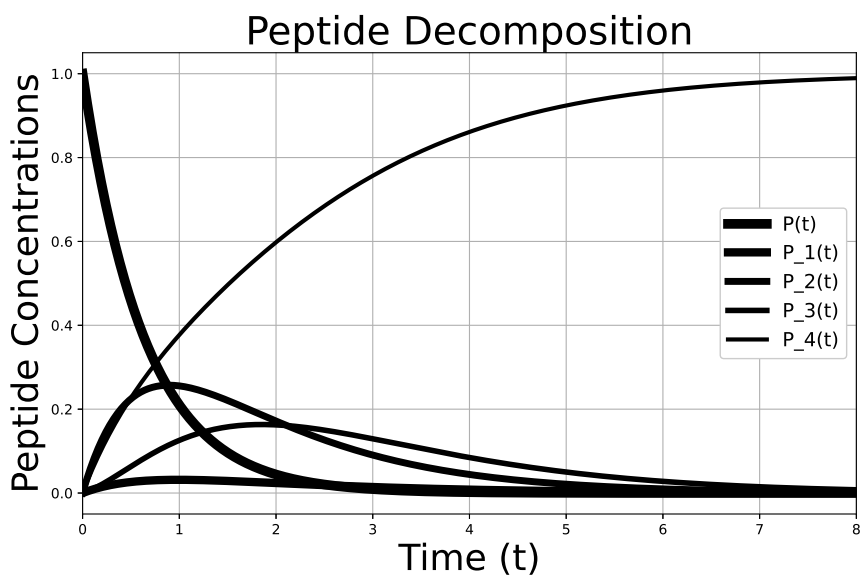


Figure 3: A time series plot of each peptide component. As time $t \to \infty$, larger peptide components eventually break down to smaller and smaller components until they all are identified as P_4. Therefore, $P_4 \to 1$ and $P.P_1, P_2, P_3 \to 0$.